

PIACENZA SAN DAMIANO, Italy

WMO: 160840

Lat: 44.913N

Lon: 9.723E

Elev: 139

StdP: 99.67

Time Zone: 1.00 (EUC)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-4.0	-11.7	1.4	3.0	-8.8	1.8	2.8	8.3	7.8	7.3	7.0	2.2	250	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	33.4	23.0	32.1	22.6	30.8	21.9	25.3	30.0	24.1	28.9	23.1	28.3	2.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.2	27.9	22.7	17.7	26.6	21.7	16.6	25.4	78.0	30.1	73.1	28.9	69.2	28.0	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.4	5.4	-7.9	35.4	2.4	2.0	-9.7	36.8	-11.1	38.0	-12.4	39.1	-14.2	40.6		
(5)			-8.2	26.2	2.4	1.1	-9.9	27.0	-11.3	27.6	-12.6	28.2	-14.3	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	3.0	4.2	8.7	12.3	17.3	21.4	23.7	18.7	13.6	8.3	3.5	(6)	
(7)		DBStd	8.01	2.56	3.58	3.21	2.92	2.69	2.97	2.55	2.83	2.70	3.46	3.21	3.10	(7)
(8)		HDD10.0	735	217	162	63	12	0	0	0	0	9	68	203	(8)	
(9)		HDD18.3	2354	476	395	299	183	51	7	1	1	28	151	302	461	(9)
(10)		CDD10.0	1920	0	1	22	80	228	343	425	425	260	120	17	1	(10)
(11)		CDD18.3	496	0	0	0	0	21	100	167	168	38	3	0	0	(11)
(12)		CDH23.3	5028	0	0	3	11	226	998	1746	1741	285	19	0	0	(12)
(13)		CDH26.7	1869	0	0	0	1	38	341	694	742	50	2	0	0	(13)
(14)	Wind	WSAvg	1.9	1.4	1.8	2.2	2.3	2.2	2.2	2.1	2.0	1.6	1.5	1.2	(14)	
(15)	Precipitation	PrecAvg	878	58	55	63	88	71	75	33	59	86	120	121	62	(15)
(16)		PrecMax	1315	166	184	195	191	147	236	124	130	276	249	251	200	(16)
(17)		PrecMin	598	2	0	6	3	5	19	4	7	9	28	30	0	(17)
(18)		PrecStd	210	43	50	47	43	38	60	27	37	62	61	67	49	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	17.4	22.9	24.9	29.9	34.0	34.2	36.2	30.0	26.2	17.1	12.9	(19)
(20)			MCWB	7.1	10.1	12.9	14.8	19.5	21.7	24.6	22.9	21.2	17.5	12.1	9.0	(20)
(21)		2%	DB	10.1	14.0	19.6	22.8	27.5	31.9	33.0	34.1	28.1	22.2	15.0	10.2	(21)
(22)			MCWB	6.3	8.0	12.0	14.1	18.0	22.3	23.5	23.0	19.6	17.6	12.9	8.6	(22)
(23)		5%	DB	8.3	11.9	17.8	20.9	25.9	30.1	31.9	32.3	26.7	20.8	13.9	9.1	(23)
(24)			MCWB	6.2	7.6	11.0	13.7	17.6	21.7	22.8	22.5	18.9	17.0	12.4	7.9	(24)
(25)		10%	DB	7.0	9.8	15.8	19.1	24.1	28.7	30.5	30.9	24.9	18.9	12.9	8.0	(25)
(26)			MCWB	5.6	6.7	10.2	12.9	17.2	20.8	21.9	21.7	18.1	15.9	11.8	7.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.3	15.0	16.7	22.1	25.7	27.8	25.9	23.2	20.4	14.9	11.1	(27)
(28)			MCD B	10.5	15.1	20.2	21.9	26.8	29.3	32.2	31.4	27.5	22.0	15.4	11.6	(28)
(29)		2%	WB	8.1	9.6	13.2	15.6	20.4	23.7	25.8	24.7	21.7	18.9	13.8	9.4	(29)
(30)			MCD B	8.7	12.8	17.3	19.8	24.2	28.3	30.2	30.2	25.5	20.5	14.5	9.8	(30)
(31)		5%	WB	6.9	8.2	11.9	14.6	19.2	22.7	24.6	23.7	20.6	17.9	12.9	8.2	(31)
(32)			MCD B	7.6	10.0	16.1	18.8	23.5	27.5	28.8	29.3	24.1	20.0	13.5	8.6	(32)
(33)		10%	WB	5.8	7.3	10.8	13.8	18.1	21.9	23.2	23.0	19.6	16.8	11.9	7.1	(33)
(34)			MCD B	6.5	8.7	14.6	17.8	22.5	26.8	28.1	28.6	22.7	18.6	12.6	7.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	8.7	11.2	10.8	11.4	11.7	12.0	11.7	10.6	7.8	5.9	5.9	(35)
(36)			MCDBR	9.5	13.0	14.8	14.3	14.0	13.6	13.0	14.0	12.9	10.1	7.6	7.8	(36)
(37)			MCWBR	6.7	8.1	8.1	7.2	6.6	5.8	5.4	5.2	5.6	5.8	5.4	5.7	(37)
(38)		5% WB	MCDBR	7.9	10.4	12.7	11.9	12.1	12.0	11.6	12.4	10.6	8.5	5.4	5.7	(38)
(39)			MCWBR	6.3	7.3	7.9	6.9	7.1	6.3	5.8	5.8	5.9	5.7	4.5	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.381	0.428	0.459	0.454	0.476	0.460	0.450	0.440	0.430	0.381	0.347	(40)	
(41)		taud	2.353	2.246	2.121	2.067	2.126	2.116	2.170	2.210	2.204	2.185	2.283	2.383	(41)	
(42)		Ebn at Noon	742	784	793	802	820	802	810	803	773	715	693	708	(42)	
(43)		Edh at Noon	80	105	136	156	151	154	144	133	124	111	85	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.47	2.25	3.56	4.57	5.67	6.53	6.72	5.72	4.25	2.51	1.50	1.24	(44)	
(45)		RadStd	0.20	0.35	0.49	0.51	0.51	0.37	0.38	0.36	0.28	0.38	0.23	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.33	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	+63	+8

Nomenclature: See separate page